

3, 5, 10, 12, 14

19

- 5.
- (1) $\{N(t) < n\} = \{W_n > t\}$
- (2) $\{N(t) > n\} \subseteq \{W_n < t\}$
- (3) $\{N(t) \leq n\} \supseteq \{W_n \geq t\}$
- (4) $\{N(t) \geq n\} = \{W_n \leq t\}$

5.

$$E[N(t)] = t\lambda$$

$$D[N(t)] = t\lambda$$

$$\begin{aligned} \mu_{N(t)} &= E[N(t)] - tE[N'(t)] \\ &= t\lambda - t \cdot \lambda \\ &= 0 \end{aligned}$$

$$\begin{aligned} D[X(t)] &= D[N(t) - N'(t)] \\ &= D[N(t)] + D[N'(t)] \\ &= t\lambda + (t-1)\lambda \\ &= t(t-1)\lambda \end{aligned}$$

5.

3.

(1) $P(N(3) - N'(1) \geq 2)$

$$= 1 - (1 + 2\lambda)e^{-2\lambda}$$

(2) $P(N(3) \geq 2 | N'(1) = 1)$

$$= P(N(3) - N'(1) \geq 1)$$

$$= 1 - e^{-2\lambda}$$

(3) $P(N'(1) = 1 | N(3) \geq 2)$

$$= \frac{P(N'(1) = 1) P(N(3) - N'(1) \geq 1)}{P(N(3) \geq 2)}$$

$$= \frac{\lambda e^{-\lambda} (1 - e^{-2\lambda})}{1 - (1 + 3\lambda)e^{-3\lambda}}$$

$$\begin{aligned} R_x(s, t) &= P_N(\min(s, t)) + s t \lambda^2 \\ x(t) &= [N(t) - N'(t)] + [(t-1)N'(t)] = A(t) + B(t) \\ R_x(s, t) &= R_A(s, t) + R_B(s, t) + \mu_A(s)\mu_B(t) + \mu_A(t)\mu_B(s) \\ &= 2(s-1)(t-1)\lambda^2 + \lambda[t(s-1) + \min\{t, s-1\}] - 2\lambda^2(s-1)(t-1) \\ &= \begin{cases} \lambda s(t-1) & 0 < s < t \leq 1 \\ \lambda t(s-1) & 0 < t < s \leq 1 \end{cases} \end{aligned}$$

(1) $P(N(s) = k | N(t) = n) = C_n^k \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k}$

(2) $P(W_3 \leq 3 | W_1 = 1) = \frac{P(T_1 = 1, T_2 \leq 2)}{P(T_1 = 1)} = 1 - e^{-2\lambda}$

(3) $P(W_k \leq s | N(t) = n)$

$$= \sum_{i=k}^n P(N(s) = i | N(t) = n)$$

$$= \sum_{i=k}^n C_n^i \left(\frac{s}{t}\right)^i \left(1 - \frac{s}{t}\right)^{n-i}$$

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{\infty} + \frac{1}{\infty} = 0$

$R_{eq} = \infty$

12. $1h$ 为单位时间, $\lambda = 6$
 $t: 00$ 为 0 时刻

$$\begin{aligned} (1) \quad & P(N_1(0.5) \geq 3) \\ &= 1 - (1 + 3 + \frac{9}{2}) e^{-3} \\ &= 1 - \frac{17}{2} e^{-3} \end{aligned}$$

$$\begin{aligned} (2) \quad & P(N_1(0.5) + N_2(0.5) \geq 3) \\ &= 1 - (1 + 6 + 18) e^{-6} \\ &= 1 - 25 e^{-6} \end{aligned}$$

$$\begin{aligned} (3) \quad & P[N_1(0.5) + N_2(0.5) \geq 6] \\ &= 1 - (1 + 6 + \frac{6^2}{2!} + \frac{6^3}{3!} + \frac{6^4}{4!} + \frac{6^5}{5!}) e^{-6} \\ &= 1 - 179.8 e^{-6} \end{aligned}$$

14.

$$\begin{aligned} (1) \quad & N_1(1) \sim \pi(2) \quad N_2(1) \sim \pi(3) \\ & P(N_1(1)=0, N_2(1)=1) \\ &= e^{-2} \cdot 3e^{-3} \\ &= 3e^{-5} \end{aligned}$$

$$\begin{aligned} (2) \quad & N_1(1) + N_2(1) \sim \pi(5) \\ & P(N_1(1) + N_2(1) = 2) \\ &= \frac{5^2}{2} e^{-5} \\ &= \frac{25}{2} e^{-5} \end{aligned}$$

$$\begin{aligned} (3) \quad & N(2) = N_1(2) + N_2(2) \sim \pi(10) \\ & P(A) = P[N(2)=3] \cdot \begin{pmatrix} C_3^1 & 0.1 & 0.9^2 \end{pmatrix} \\ &= \frac{1}{3} \frac{81}{2} e^{-10} \end{aligned}$$

(4)

垃圾邮箱数目在 $(0, t]$
 内是强度为 $5 \times 0.1 \times t = \frac{t}{2}$
 泊松分布, 记为 $A(t) \sim \pi(\frac{t}{2})$

$$\begin{aligned} & P(A(1)) \\ &= P(1 < W_2 \leq 2) \\ &= P(A(1)=0) P(A(2)-A(1) \geq 2) + P(A(1)=1) P(A(2)-A(1) \geq 1) \\ &= \frac{3}{2} e^{-\frac{1}{2}} - 2e^{-1} \end{aligned}$$